Challenging Problems

6. Three consecutive integers, when taken in increasing order and multiplied by 2, 3 and 4 respectively, total up to 74. Find the three numbers.
7. 331 students went on a field trip. Six buses were filled to capacity and 7 students had to travel in a van. How many students were there in each bus?
8. A mobile vendor has 22 items, some which are pencils and others are ball pens. On a particular day, he is able to sell the pencils and ball pens. Pencils are sold for ₹15 each and ball pens are sold at ₹20 each. If the total sale amount with the vendor is ₹380, how many pencils did he sell?
9. Draw the graph of the lines $y = x$, $y = 2x$, $y = 3x$ and $y = 5x$ on the same graph sheet. Is there anything special that you find in these graphs?
10. Consider the number of angles of a convex polygon and the number of sides of that polygon. Tabulate as follows:

Name of Polygon	*No. of angles*	*No. of sides*

Use this to draw a graph illustrating the relationship between the number of angles and the number of sides of a polygon.

SUMMARY

- When the product of two algebraic expressions we follow,
  Multiply the signs of the terms,
  Multiply the corresponding co-efficients of the terms.
  Multiply the variable factors by using laws of exponents.
- For dividing a polynomial by a monomial, divide each term of the polynomial by a monomial.
- Identity: An identity is an equation is satisfied by any value that replaces its variables (s).

$(a+b)^2 = a^2 + 2ab + b^2$ $\quad$ $(x+a)(x+b) = x^2 + (a+b)x + ab$

$(a-b)^2 = a^2 - 2ab + b^2$ $\quad$ $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

$a^2 - b^2 = (a+b)(a-b)$ $\quad$ $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

$(x+b)(x+b)(x+c) = x^3 + (a+b+c)x^2 + (ab+bc+ca)x + abc$

- **Factorisation:** Expressing an algebraic expression as the product of two or more expression is called Factorisation.
- An equation containing only one variable with its highest power as one is called a linear equation in one variable.
- The value which replaces a variable in an equation so as to make the two sides of the equation equal is called a solution or root of the equation.
- Graphing is just a visual method for showing relationships between numbers.
- The horizontal line is named as XOX', called the X-axis. The vertical line is named as YOY', called the Y-axis. Both the axes are called **coordinate axes**. The plane containing the x axis and the y axis is known as the coordinate plane or the **Cartesian plane**.
- A point is denoted by a pair (a,b) of two numbers 'a' and 'b' listed in a specific order in which 'a' represents the distance along the X-axis and 'b' represents the distance along the Y axis. It is called an ordered pair (a,b).
- The coordinate axes divide the plane of the graph into four regions called quadrants.
- The line graph for the linear equation is called a **linear graph**.

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